RC Low-Pass Filter: Step Response and Frequency Response in LTspice

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1 Introduction

In this report, I study a first-order RC low-pass circuit with $R = 10\,\mathrm{k\Omega}$ and $C = 100\,\mathrm{nF}$. I derive its transfer function with the Laplace transform, compute its step response and its cutoff frequency by hand, and then verify the results with LTspice transient and AC simulations.

2 Theory

2.1 Transfer function

Replacing the capacitor by its impedance $Z_C = 1/(sC)$, the circuit is a voltage divider:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1/(sC)}{R + 1/(sC)} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau}$$

The time constant is

$$\tau = RC = 10\,\mathrm{k\Omega} \times 100\,\mathrm{nF} = 1\,\mathrm{ms}$$

so the pole is at

$$s = -\frac{1}{\tau} = -1000\,\mathrm{rad/s}$$

which corresponds to the cutoff frequency

$$f_c = \frac{1}{2\pi RC} = \frac{1000}{2\pi} \approx 159.15\,\mathrm{Hz}$$

2.2 Step response

For a $5\,\mathrm{V}$ step, $V_{in}(s) = 5/s$, so

$$V_{out}(s) = \frac{5}{s\,(1 + 0.001\,s)} = \frac{5000}{s\,(s + 1000)} = \frac{5}{s} - \frac{5}{s + 1000}$$

and the inverse Laplace transform gives

$$v(t) = 5 - 5\,e^{-1000\,t}\ \text{V}$$

Checks: $v(0) = 0$ and $v(\infty) = 5\,\text{V}$, as expected for an initially empty capacitor.

Time	$v(t)$
0	0 V
1 ms (τ)	3.16 V
5 ms (5τ)	4.97 V

Table 1: Hand-calculated capacitor voltage

2.3 Frequency response

Setting $s = j\omega$:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}, \qquad \angle H(j\omega) = -\arctan(\omega\tau)$$

At f_c the magnitude is $1/\sqrt{2}$ ($-3\,\text{dB}$) and the phase is $-45°$. At 50 Hz:

$$\omega = 2\pi \cdot 50 \approx 314.2\,\text{rad/s}, \quad \omega\tau \approx 0.314, \quad |H| = \frac{1}{\sqrt{1 + 0.0987}} \approx 0.954$$

$$20\log_{10}(0.954) \approx -0.41\,\text{dB}$$

Mains noise at 50 Hz is therefore almost not attenuated by this filter.

3 LTspice Simulation

3.1 Transient analysis

The source V1 is a pulse source, `PULSE(0 5 0 1n 1n 5m 10m)`, with the command `.tran 10m`. The capacitor voltage is taken at the node between R1 and C1.

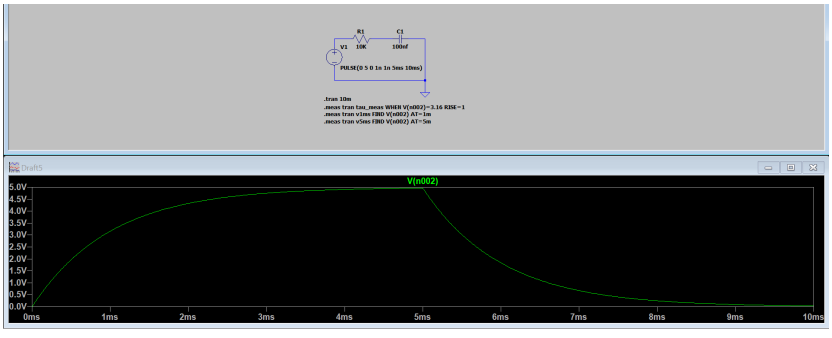


Figure 1: Capacitor voltage: charging and discharging

3.2 AC analysis

The AC amplitude of V1 is 1 V, with the command `.ac dec 100 1 100k`. The cutoff frequency was measured with `.meas ac fc_meas WHEN mag(V(n002))=0.7071 FALL=1`.

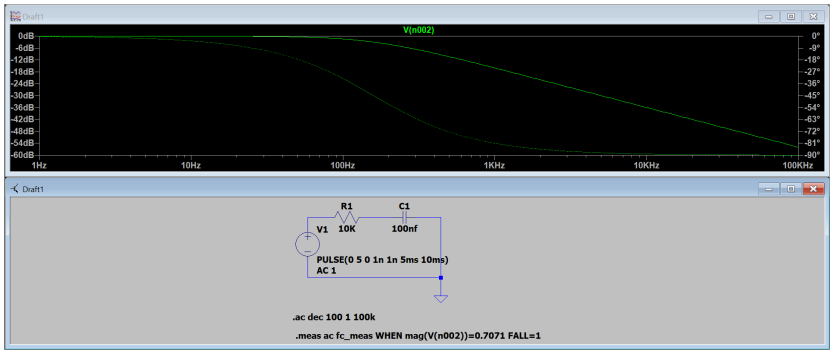


Figure 2: Bode plot: magnitude (solid) and phase (dotted)

4 Results

Quantity	Hand calculation	LTspice
τ	1 ms	0.99967 ms
f_c	159.155 Hz	159.158 Hz
$v(1\,\text{ms})$	3.16 V	3.1606 V
$v(5\,\text{ms})$	4.97 V	4.9663 V
$\lvert H\rvert$ at 50 Hz	−0.41 dB	−0.4088 dB

Table 2: Hand calculation compared with simulation

The measured cutoff frequency matches the calculated one to within 0.003 Hz. The small difference comes from the rounded value 0.7071 used in the measurement instead of $1/\sqrt{2}$.

5 What I Learned

- A missing unit letter changes a value by orders of magnitude: writing `1` instead of `1n` in the pulse source made the rise time one second, and the capacitor followed the source instantly.
- The simulation stop time must match the circuit time scale: `.tran 10` showed 1000 pulses squeezed into one plot.
- In LTspice, clicking a component plots its current, and clicking a wire plots its voltage.

- Checking a solution at $t = 0$ and $t \to \infty$ quickly reveals errors in the Laplace inverse transform.
- With $f_c = 159\,\text{Hz}$, a $50\,\text{Hz}$ signal is hardly attenuated, so a lower cutoff frequency is needed to remove mains noise.